Muhammad Arslan Aslam

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WORK EXPERIENCE

01/01/2024 – CURRENT Leipzig, **Germany**

RESEARCHER HELMHOLTZ CENTRE FOR ENVIRONMENTAL RESEARCH

DiPEM Project (Digital Molecular Endocrine Profiling)

1. Conducting **zebrafish embryo (ZFE) toxicity assays** to assess the effects of **endocrine-disrupting chemicals (EDCs)**.
2. Contributing to the development of mechanistically anchored **digital biomarkers** by linking phenotypic effects with **AOPs** to expand and refine the Fish Early Life Stage Test (FELS).
3. Supporting the replacement of adult-based FSTRA with embryo-based **NAMs**, aligning with 3Rs and OECD guidance on endocrine disruptor testing.

EU PrecisionTox Project (Mechanistic Toxicology Using New Approach Methods (NAMs))

1. Contributed to **high-throughput phenotypic** screening of 200 chemicals using zebrafish embryos (ZFE) to assess adverse outcomes relevant to human health.
2. Employed automated image-based analyses to evaluate **morphological, cardiovascular, and behavioral endpoints** (e.g., photomotor response, spontaneous movement), including those potentially linked to neuro- and endocrine activity.
3. Assessed specificity of chemical effects by calculating sensitivity ratios (**predicted baseline EC50** / measured EC50)
4. Used effect specificity patterns to support **mode-of-action (MoA)** inference and improve predictive toxicology for structurally uncharacterized compounds especially endocrines.

EU PARC Project (Partnership for the Assessment of Risks from Chemicals)

1. Assessed endocrine-disruptive potential of **BPA** and four structural analogues (BPE, BPAP, BPZ, BPG) by morphological, cardiovascular, and behavioral effects of zebrafish embryos (ZFE).
2. Developed novel phenotypic endpoints and trained **Python-based models** for automated annotation of morphology.
3. Applied PrecisionTox specificity framework (baseline EC50 / observed EC50 >10) to identify specific-acting compounds; BPE showed the highest specificity and overall toxicity.

01/01/2023 – 30/06/2023 Bilbao, **Spain**

RESEARCH INTERNEE UPV/EHU PIE MARINE STATION

1. Prepared Water Accommodated Fractions using heavy and light oils for **Sea urchin embryo toxicity**
2. Significant reductions in fertilization success and larval development observed
3. Identifying and quantifying chemicals using **LC-HRMS**

EDUCATION AND TRAINING

01/09/2022 – 01/08/2024 Europe

ERASMUS MUNDUS MSC IN MARINE ENVIRONMENT AND RESOURCES

Co-funded by the Erasmus+ Programme of the European Union

1. University of Bordeaux, France **(1st Semester)**
2. University of the Basque Country, Spain - UPV/EHU **(2nd Semester)**
3. University of Liege, Belgium **(3rd Semester)**
4. Helmholtz Centre for Environmental Research**,** Germany **(Master Thesis)**

Thesis: COMPARATIVE TOXICITY ASSESSMENT OF BISPHENOLS ADDITIVES USING A HIGH CONTENT ZEBRAFISH ASSAY

1. This study evaluated the developmental toxicity of BPA and its substitutes (BPE, BPAP, BPZ, BPG) in zebrafish embryos using multi-endpoint, high-content screening approach.

2. The results demonstrated that BPE exhibited the highest toxicity, affecting multiple endpoints, including neurological and cardiovascular development.

3. This research provides important insights for the regulatory assessment of bisphenol substitutes and their potential health risks.

BACHELOR IN ENVIRONMENTAL SCIENCES - 3.84/4 (BRONZE MEDALIST) Government College University Faisalabad

Thesis: Awareness, Disposal Behaviors, Willingness, and Inflow of Electronic Waste in Faisalabad, Pakistan

1. Designed and conducted a mixed-methods study (surveys, field visits, and questionnaires)
2. 75.2% of respondents were unaware of e-waste; 82% lacked knowledge of its hazards; 45% relied on informal recycling; 91% showed willingness to participate in formal management schemes.
3. Contributed to sustainable waste management discourse by providing data-driven evidence for targeted interventions and policy development in Pakistan.

SPECIALIZATION IN CONCEPTS OF TOXICOLOGY AND CHEMICALS Jhons Hopkins University

Toxicology 21: Scientific Applications
- Concepts of Pathways of Toxicity (PoT)

Chemicals and Health
- Risk Assessment of Chemicals
- Regulatory Framework and

Evidence-based Toxicology
- Systematic Review and Meta-Analysis
- Biases in Conducting Toxicity Assays

DIGITAL SKILLS

Experience in python, and R | Reference Management in Zotero, Mendeley | Software Tools: KNIME and other common applications (e.g. Microsoft Office, LaTeX)

	R	Python	KNIME
Data Processing	●●●●●	●●●○○	●●●●●
Statistical Analysis	●●●●○	●●●○○	●●●●○
Visualization	●●●●●	●●●●○	●●●●●
Machine Learning	●●●○○	●●○○○	●●○○○

PUBLICATIONS

Research Article(s)

Alghanem, S. M. S., Alsudays, I. M., Farid, M., Sarfraz, W., Ishaq, H. K., Farid, S., Zubair, M., Khalid, N., **Aslam, M. A.**, Abbas, M., & Abeed, A. H. A. (2024). **Evaluation of heavy metal accumulation and tolerance in oxalic acid-treated Phragmites australis wetlands for textile effluent remediation**. International Journal of Phytoremediation, 1–16. https://doi.org/10.1080/15226514.2024.2372849

Aslam, M. A., Ahmad, M. M., Farid, S., & Hashmi, S. I. (2025). **Awareness, disposal behaviours, willingness, and inflow of electronic waste in Faisalabad, Pakistan.** International Journal of Environment and Waste Management, 36(4), 472–490. https://doi.org/10.1504/ijewm.2025.144994

Book Chapter(s)

Salimkumar A.J., Cleetus M.C.K., Essel D.A., **Aslam M.A.,** Galib A.H., Prabhakaran M.P., and Rejish K.V.J (2024) **Plastic Peril: Unraveling Microplastics Threats to Health and Ecosystems**. In: Singh N.K., Jayaswal K., Yadav M., Maharjan N., editors. Fate of Microplastics in Wastewater Treatment Plants Occurrence, Identification, Potential Factors, and Future Perspectives. 1st ed. Tylor & Francis: CRC Press (22 November, 2024). https://www.routledge.com/Fate-of-Microplastics-in-Wastewater-Treatment-Plants-Occurrence-Identification-Potential-Factors-and-Future-Perspectives/Singh-Jayaswal-Yadav-Maharjan/p/book/9781032538778

Farid, M., Ahsan, A., Asam, Z. U. Z., Abbas, M., Fatima, A., Salman, M., Tariq, S., Iqbal, A., Akhtar, W., Ali, S., & **Aslam, M**. (2023). **Principles and Applications of Environmental Biotechnology for Sustainable Future.** In Springer eBooks (pp. 751–775). https://doi.org/10.1007/978-3-031-37424-1_34

Farid, M., Farid, S., **Aslam, M.**, Azhar, K., Tariq, K., Ali, Z., Ijaz, Z., Sarfraz, Z., Khalid, M., & Ahmad, M. (2023). **Biotechnological techniques for sustainable waste management.** In Springer eBooks (pp. 689–712). https:// doi.org/10.1007/978-3-031-37428-9_30

HONOURS AND AWARDS

Her Majesty Queens Thailand Scholarship – Asian institute of Technology, Thailand

Award of Laptop on Securing 2nd position in university – Ministry of Education, Pakistan

European Erasmus Mundus Scholarship –

CONFERENCES & SEMINARS

Presenter, PrecisionTox Annual Consortium Meeting, Brussels, Belgium, June 13–14, 2024.

Poster Presenter, PARC Consortium Meeting, Hall in Tirol, Austria, May 13–16, 2024.

Project Presenter, Asian Institute of Technology Sustainability Hackathon 2021

RECOMMENDATIONS

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